

CSA1017 – SOFTWARE ENGINEERING

INDIVIDUAL REFLECTION REPORT

INTELLIGENT DISASTER EARLY WARNING AND EVACUATION MANAGEMENT SYSTEM

NAME: LOSHINI L

REGISTER NO.: 192571015

ROLE: Evacuation Route and Shelter Management

DESIGN / DEVELOPMENT DECISION:

I contributed to the evacuation and shelter-management component of the system. My focus was on helping users identify suitable evacuation routes and nearby emergency shelters during cyclone-induced flooding. I considered important information such as affected roads, shelter locations, available capacity, and safer areas.

CHALLENGE FACED AND SOLUTION:

The main challenge was ensuring that evacuation information remains simple and useful during an emergency. I addressed this by focusing on clear route information, shelter details, and emergency instructions. The design prioritises the information that citizens need immediately during evacuation.

LEARNING GAINED:

I learned how software systems can support emergency planning and evacuation management. I understood that evacuation systems need accurate location information, simple instructions, and timely updates because conditions can change rapidly during disasters.

SDG RELEVANCE:

The project supports SDG 11 by improving disaster preparedness, evacuation planning, and community resilience. Providing accessible shelter and route information can help reduce risks to citizens.

COURSE OUTCOME CONTRIBUTION:

My contribution supports course outcomes related to requirements engineering, system design, usability, GIS integration, and emergency workflow development. I learned how software requirements must reflect real-world emergency situations.

POSSIBLE IMPROVEMENT:

I would improve the evacuation module by adding dynamic route updates, shelter-capacity tracking, traffic information, accessibility-friendly routes, offline navigation, and automatic rerouting when roads become unsafe.